# Phenotypic Plasticity

When Elina and Nori venture into the Deapths of Despair to save Nalu, Elina's wings do not give her enough strength to propel herself through the water as effectively as Nori can with her tail. In order to swim against the strong current and help Nori, she had to trade her wings for a tail and undergo Morphological Remodeling.

Phenotypic plasticity is an organism's ability to change its physical traits (morphology), functions (physiology), or behaviors in response to environmental cues. This allows a single genotype to produce different phenotypes—observable traits that shift under varying environmental conditions. Much like a caterpillar mimicking a flower or a twig in response to internal or external signals (e.g., light, temperature, food quality, predator presence).

The source of these transformations varies based on the available biological resources and the immediacy of the environmental demand. Initially, Elina's metamorphosis is mediated by her necklace, acting as an external prosthetic cue that temporarily overrides her default morphology. However, her final transition is triggered by the True Self Berry, a biochemical catalyst that induces a permanent, internal shift in gene expression.

This demonstrates that phenotypic plasticity is not a singular event, but a flexible response system: whether through a temporary environmental signal or a profound metabolic ingestion, the organism reallocates its physical 'self' to match the constraints of its reality.

# 2005 Pandemic

## World of Warcraft Corrupted Blood Incident

The Corrupted Blood Incident (CBI) was an accidental virtual epidemic that occurred in the MMORPG World of Warcraft (WoW) developed by Blizzard Entertainment between September 13 and October 8, 2005. The event is significant because it provided epidemiologists with a unique "human-agent lab" to study disease transmission and social behavior during a crisis. It started in newly added raid Zul'Gurub.

The final boss of Zul'Gurub, Hakkar the Soulflayer, cast the debuff Corrupted Blood on raid participants, which was supposed to expire when players defeated Hakkar. The debuff was meant to force coordination, when a player contracted it, the countdown would reset, and without distancing, it would spread further. The breach comes down to hunters and warlocks - the classes that use pets. The players would deactivate their infected animal companions, which would freeze their state. Virus in cryostasis. When back from the raid, in densely populated non-combat zones, like an an auction house, they would summon their pets, and the debuff would spread again.

raid = team of players (~20) organized like a sports team, going through a battle scenario, to gain experience and rewards.

**Corrupted Blood**

Deals 263 to 337 damage every second to the target and all nearby allies for 10 sec. Spreads to nearby allies

The most critical point was NPCs contracting the debuff. The pets were disease vectors, while non-player characters became asymptomatic carriers. They have a massive health bar and cannot be 'just killed'.

One of the authors of the cited paper witnessed the outbreak personally. He was 10 years old at that time, he logged in, and saw the streets paved with skeletons. He was genuinely scared. The thought that the major Horde city Orgrimmar was under the attack of the opposing faction, or that it was being attacked by a world boss.

This collapse of the “safe zone” led to genuine feelings of fear and confusion among players, blurring the line between the fictional game world and players' emotional responses.This is described as the collapse of the 'magic circle' - the conceptual boundary that separates play from reality, giving new layer to moral decisions.

Player responses varied. Some acted as real-world health workers, trying to heal infected players. Some spread the disease intentionally, either out of curiosity or malicious intent. There were also bystanders who observed the incident without actively participating. Morality in games becomes optional. But it still mirrors what we see in real outbreaks - panic, flight, altruism, fear, and malice. Some players fled the cities, self quarantined, or entirely logged off.

There was also a huge misinformation problem, Blizzard stayed silent because they scrambled to fix to code, they didn't have a PR statement for accidental digital plague, so the public chat was on fire with theories of what's going on. Blizzard did a server reset to wipe the pandemic.

Once the dust settled the researchers started looking at what just happened. It was a simulation they could never run legally. A new methodology introduced itself. Computer models are usually used to calculate and predict the course of disease outbreaks, which is static, with no real human behavior. The math of the virus collided with real unpredictable nature of humans.

However it is still an online game and the risk taking is skewed. Dying is a mere inconvenience, you have to run back to your body, lose some money on repairs, but the game isn't over. The existence of hardcore servers could counter this, at least partially. When a player dies, their character is deleted. The panic and risk taking would be very much realistic. But on the contrary, the motivation to spread the viruses would have been much stronger.

# Natural Kevlar

**Silk** is composed primarily of a protein called **Fibroin**. At the microscopic level, fibroin consists of tightly packed, flat "sheets" of proteins. These are called **Antiparallel Beta-Pleated Sheets**. These sheets are held together by a massive density of Hydrogen Bonds. Individually, a hydrogen bond is weak, but millions of them working together act like molecular "**Velcro**."

## Flowy Armor

Between these crystalline "hard" sheets are amorphous, "stretchy" regions. This combination is what gives silk its unique profile: the crystals provide **tensile strength**, while the amorphous regions provide **elasticity**.

## Balistic Properties

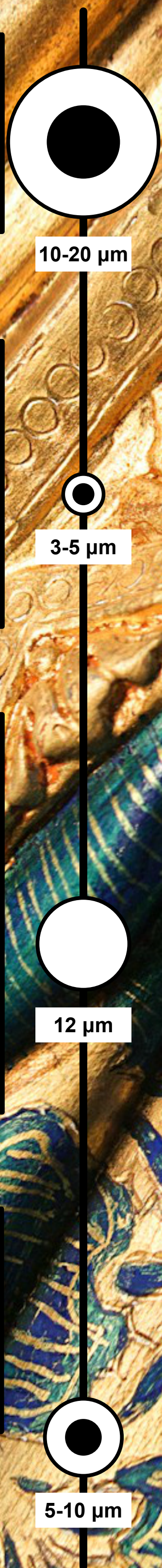
Historically, silk was used in early body armor. Because it is so strong, a silk vest could actually stop low-velocity black-powder bullets or arrows. Instead of the projectile piercing the skin and shattering bone, the silk would "catch" the bullet and pull it into the wound as an intact bag, making it much easier to remove and preventing infection.

## The Shield of Brocade

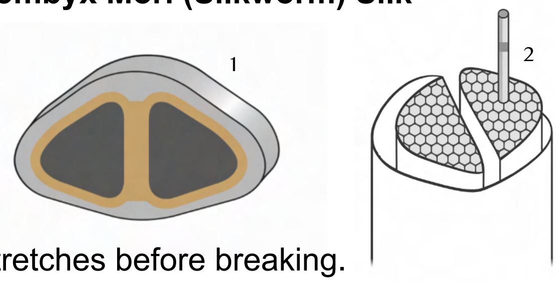
Natural silk fibers are built like a high-tech cable, consisting of a core of strong protein threads called fibroin held together by a sticky, protective "glue" known as sericin. At the microscopic level, these fibroin threads are made of even smaller bundles of nanofibrils, where rigid crystals provide the strength while flexible chains allow for stretch. This unique combination of a "tough core" and a "shielding skin" is what allows silk to be incredibly lightweight yet stronger than steel by weight.

## The Shield of Brocade

In the courts of Europe or the Edo period in Japan, wearing heavy Brocade wasn't just about fashion. The sheer cost of the textile signaled a level of invulnerability. Being draped in heavy silk Brocade, made its bearer physically and legally untouchable.

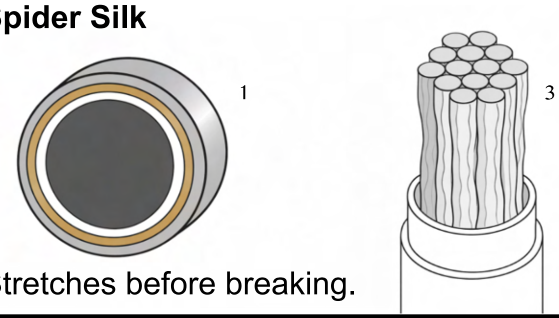



## Bombyx Mori (Silkworm) Silk



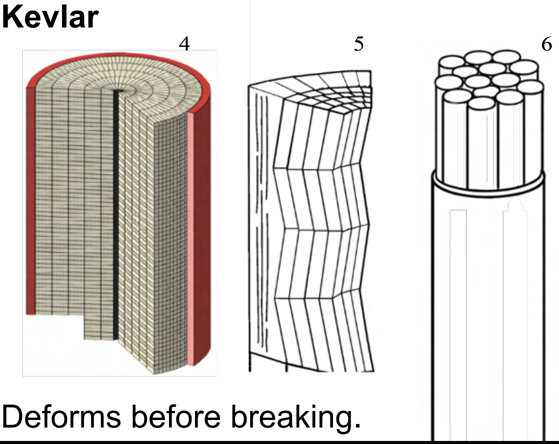

Stretches before breaking.

## Spider Silk



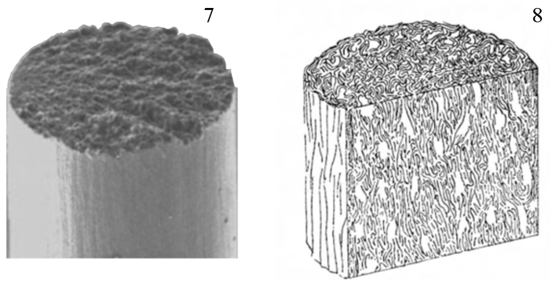

Stretches before breaking.

## Kevlar




Deforms before breaking.

## Carbon



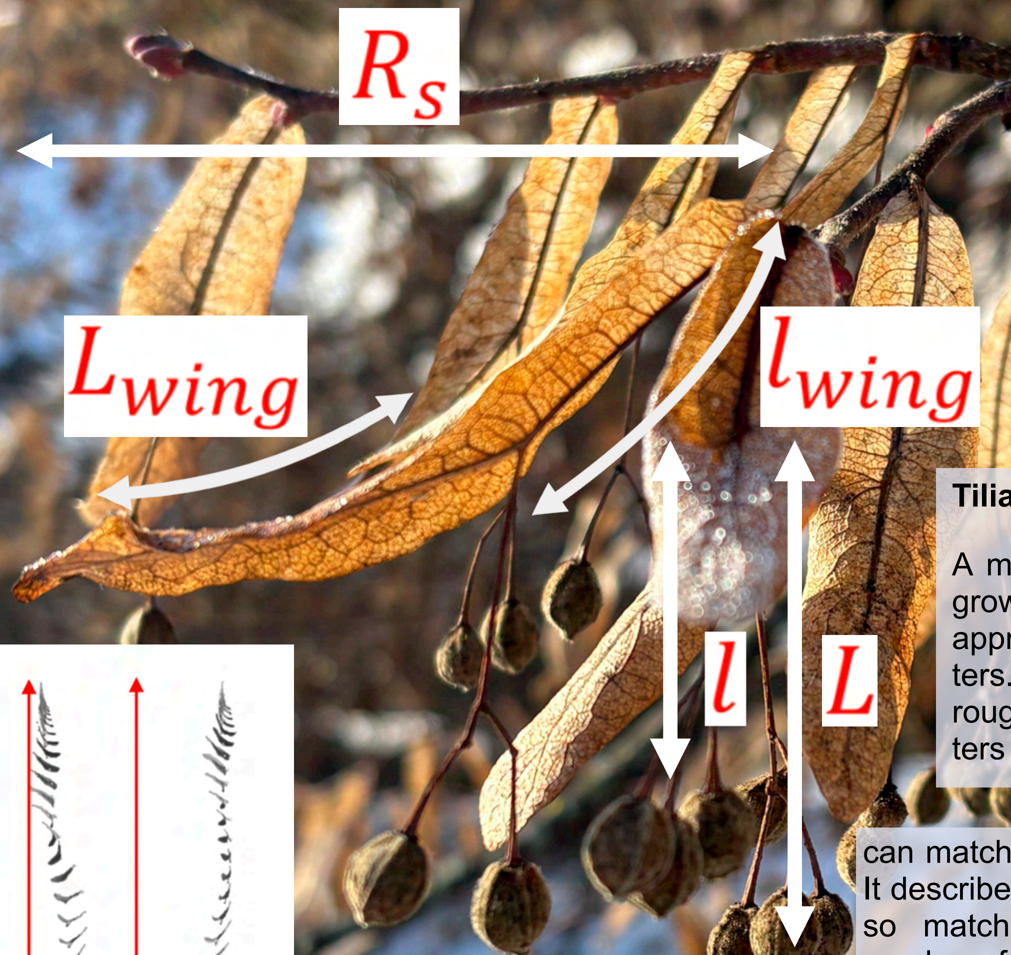

Doesn't stretch or deform, shatters under impact.

# Bract Monopteron

The seed needs to be carried away as far as possible from the tree, so that it doesn’t land in its shadow or on its roots, where it would be starved of nutrients. The wind is fickle, you can’t schedule a gust, the only variable the tree can control is the time. When you drop a rock from 30 meters it hits the ground in around 2,5 seconds. The seed needs to fall as slow as possible.

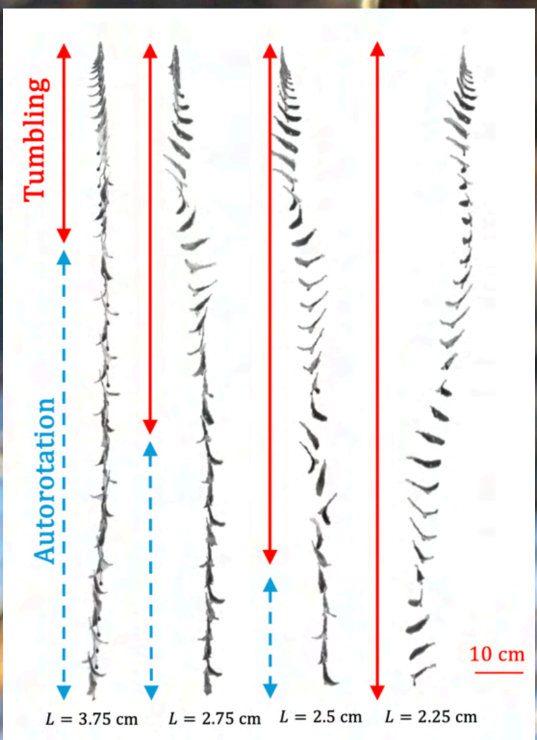
There is tumbling and autorotation. Tumbling is chaotic movement where gravity wins quickly. Autorotation is steady, stable rotating motion around the vertical axis - the wing is generating lift, turning the fall into a glide. Autorotation cuts the descend speed in half. Doubling the dispersed range is in evolutionary terms superpower.

The length of the stalk matters, when it falls from the tree it doesn’t start spinning immediately, it tumbles for a split second. Then it has to self-correct and lock into the spin. The stalk is what makes this correction possible. The movement of the seed creates torque through the stalk, which pulls the wing back into the correct orientation. Shorter stalk would result in less torque, and there would be no correction, and no gliding.


**Tilia x europaea**

A mature tree typically grows to a height of approximately 30 meters. Its crown begins roughly two to five meters above the ground.




Scientists 3D printed perfect replicas of the linden seeds, took a tank of water and filled it with tiny glittering particles, to be able to see the water moving, by shining a laser at the water.

The only difference between air and water is density and viscosity. Adjusting the scale of the model and the speed of falling can match Reynolds number. It describes how a fluid flows, so matching the Reynolds number of water and air gives the same descend in both fluids.

The curvature of the wing pushes the vortex outward. The curve makes the spin stronger. Flatter wing wouldn’t need as long stalk, shorter stalk would make the curved wing tumble. Beautiful holistic system.

# Sentient Ore

In the frozen wastes of Northrend, the frozen land is corrupted by Saronite, a teal-colored metal that pulses with a faint, maddening light. To the races of Azeroth, it is the crystallized blood of the Old God Yogg-Saron, the "Lucid Dream," an entity of pure madness imprisoned beneath the earth. To a scientist, however, Saronite represents a catastrophic convergence of forced biomineralization, molecular mimicry, and acidification.

Just as Saronite is the physical manifestation of an entity's blood, real-world minerals often form through biological interactions. Hematite (Fe2O3), an iron oxide, can be formed through microbial-induced weathering and precipitation. Similarly, Cinnabar (mercury sulfide or HgS) is a deep red ore historically used for pigments, paralleling the "blood" aesthetic.

From a biogeochemical perspective, Saronite could be viewed as a result of "forced biomineralization." In extreme environments, organisms often sequester toxic metal ions into mineral structures to ensure survival. For instance, certain bacteria can precipitate toxic gold complexes into "nano-nuggets" on their cell walls. If we view Yogg-Saron’s corruption as a biological force, Saronite functions similarly to how terrestrial microbes transform iron and sulfur into minerals like greigite or pyrite.

The most insidious trait of Saronite is its ability to drive miners and smiths to madness through "whispers." In toxicology, this psychological deterioration mirrors the effects of Chronic Mercury Poisoning, specifically the neurological disturbances associated with elemental and organic mercury.

Historical exposure to mercury vapors in hat-making industries led to "Mad Hatter" syndrome, characterized by profound irritability, pathological shyness, insomnia, and psychotic behaviors. Similarly, the "Danbury shakes" described a coarse tremor interrupted by myoclonic jerks found in workers exposed to mercury vapor.

Saronite likely utilizes a mechanism known as Molecular Mimicry. In the real world, organic mercury (Methylmercury) binds with cysteine to form a complex that looks structurally identical to the essential amino acid methionine. This allows the toxin to use the L-type neutral amino acid carrier transport system—a molecular "Trojan Horse"—to trick the brain into letting it in.

Once inside the brain, these heavy metals disrupt neurotransmitters. Mercury exposure alters the metabolism of dopamine, glutamate, and GABA. This disruption leads to the "neurological disturbances" and "hallucinations" that a WoW character might interpret as the whispers of an Old God.

In the Borean Tundra and Howling Fjord, Saronite deposits are depicted as bleeding into the landscape, destroying local flora and corrupting the soil. This is a fantasy representation of Acid Mine Drainage (AMD).

When sulfide-based ores are unearthed and exposed to water and oxygen, they undergo oxidation. This chemical reaction releases ferrous iron and sulfate, generating highly acidic conditions.

As the acidity increases (pH drops), it dissolves other heavy metals from the surrounding rock, mobilizing toxins like lead, copper, and arsenic.

Just as Saronite kills flora, AMD creates an environment where most aquatic life cannot survive due to high acidity and heavy metal toxicity.

Just as the influence of the Old Gods accelerates the corruption of Azeroth, real-world bacteria act as biocatalysts. Microorganisms like Acidithiobacillus ferrooxidans thrive in these acidic environments, accelerating the oxidation of sulfide minerals and the production of acid, effectively speeding up the destruction of the ecosystem. The "plague" of undeath spread by Saronite is, geologically speaking, a runaway chemical oxidation event catalyzed by bacteria.

# Deconstructing the Sparkle

Glitter is engineered high composite material. It has a PET core, coated with a microscopic layer of metal, which is then coated with a protective plastic layer. It enters the environment through wastewater (washing off cosmetics), or via direct littering. It is frequently underestimated in scientific studies because standard detection methods often fail to capture.

## Counting the devils

When counting microplastics you would go to a beach. scoop up some sand, and you separate the plastic to count it. Or you would take a sample of ocean sediment, mix the sample with a salt solution, so that the plastic floats and the rest sinks. The density of the salt solution is 1,2g per cubic centimeter. Most common plastics are lighter than that. However, glitter sinks to the bottom.

The sediment also has to be cleaned before being analyzed, that is usually done by using some acid or chemical, but that dissolves the glitter, leaving only the PET core. Under a microscope it no longer looks like glitter but a generic plastic particle. So there is a huge data gap. The number of particles found in the ocean is actually low, when glitter isn't accounted for, and then we cannot regulate what we cannot measure.

## Alt Glitter

In response to plastic pollution concerns, alternatives such as modified regenerated cellulose (MRC), synthetic mica, and mineral-based glitters have entered the market. However, "biodegradable" does not necessarily mean environmentally safe. A study comparing conventional PET glitter to biodegradable MRC glitter found that the biodegradable options caused stronger negative effects on freshwater duckweed, including reduced root length and biomass. Alternative glitters may still contain coatings and additives that leach into the water as the particle degrades, potentially making them more chemically active in the short term than conventional plastic.

**Mica** is a common mineral alternative used in cosmetics. Elemental analysis revealed that synthetic mica glitter contained a wider range and higher quantities of metals—including aluminum, iron, manganese, and titanium—compared to PET glitter. In terms of biological toxicity to duckweed, synthetic mica showed "little effect" compared to the high toxicity of MRC. However, mica glitters can still reduce the biomass of primary producers in the water.

## Toxicity and Leaching

Sea urchins and brown mussels are studied as bioindicators. Mussels filter huge amounts of water, so when they are sick, the water is sick. Sea urchin embryos are extremely sensitive to chemical changes.

Glitter is leaching heavy metals and chemical additives such as parabens, benzene, toluene, and elemental silver - exceeding marine safety levels. Heavy metals like silver are toxic and disrupt cellular processes.

## Color Matters

Green glitter has a much more complex chemical profile than white glitter. 10mg per liter is enough to result in developmental failure of brown mussels, their embryos would fail vto reach larval state. The substances identified in glitter, such as propylparaben, are used to extend the shelf life of personal care products but are known endocrine disruptors. These particles are frequently used in direct-contact products like body paints, nail polish, and cosmetics.

## Ok Queen, Now What?

"Eco-friendly" alternatives often rely on coatings and additives to maintain their shine, which can make them chemically active and toxic in aquatic environments. Therefore, biodegradable materials are not necessarily safer than conventional plastics. When it comes to makeup and cosmetics, mica (ideally ethically sourced) is better for contact with skin, but not necessarily for the environment. You can opt for natural, organic makeup.

## Sewage Fertilizer

The solid byproduct of Sewers treatment is sewage sludge. This sludge is used as fertilizer on agricultural land (a common practice globally). Because of the density of glitter, it sinks during wastewater treatment rather than floating, and flowing out with the treated water. When the sludge is used as fertilizer, the accumulated glitter is reintroduced into the terrestrial environment. From there, it can leach chemicals or eventually wash back into waterways, creating a cycle of pollution.

## Makeup, the Better Options

Brands like Inika, Ilia, RMS, Und Gretel, ĀTHR BEAUTY, NUI Cosmetics, or (M)ANASI, use mica instead of glitter, and generally avoid microplastics, PET or PVC, microbeads. Great news, but even organic makeup needs to add some color to its natural shine, so heavy metals are oftentimes still used. Like Titanium Dioxide (CI 77891) and Iron Oxides (CI 77491) in RMS Beauty Eyelights Cream Eyeshadow Luster. Sounds bad in this context, but organic makeup is healthier than the traditional products. Just make sure to avoid propylparaben and butylated hydroxytoluene (BHT).

# Frisson

## *When you are the music*

**Aesthetic chills** are a psychophysiological response characterized by goosebumps (piloerection), shivers, and tingling sensations. They are mechanistically distinct from simple pleasure; chills represent peak **emotional arousal.** This experience is fundamentally linked to the brain's reward circuitry, specifically the release of dopamine in the striatum.

When the beat drops, or the harmony resolves—the neural activity shifts ventrally to the nucleus accumbens (NAcc). This is the moment of the chill. **Dopamine** floods the system in a way that mirrors the biological high of food, sex, or cocaine. This temporal split explains why the "chase" of the melody is just as chemically potent as the finale; the brain rewards the prediction (anticipation) differently than the outcome (consumption). Physiologically, chills are consistently associated with increased sympathetic nervous system activity, including higher skin conductance (SCR), heart rate, and respiration depth.

Two people listen to Miserere by Allegri (a validated "Gold Standard" chill-inducer). One person feels nothing. The other feels a cold shiver down their neck and arms. The difference often lies in personality. The person experiencing the chill likely scores high in **Openness to Experience**.

Specifically, this person is likely high in the **"Fantasy"** and **"Ideas"** sub-facets of Openness. They are not just feeling the music; they are cognitively immersing themselves in it, engaging in "active imagination" and rapidly processing the complex patterns. This propensity is partially written into their DNA, with twin studies suggesting the tendency to feel chills is roughly 36% heritable, and women appearing slightly more prone to reporting chills than men.

Consider a choir or band performing. There is a specific moment where the ensemble creates a series of dissonant chords that finally resolve. For the musicians, this often manifests as a feeling that the performance has "clicked" or that the group is operating as "one mind".

This is not just a metaphor; it is an intensification of communal sharing relations, often linked to the social emotion **kama muta** (being moved by love/connection). Surveyed musicians describe this as "electric," a moment of "perfection" or "alignment" that feels greater than the sum of its parts. Even if you are listening alone in your room, if you see a digital map indicating others are listening with you, the "illusory" social presence significantly boosts the pleasure and intensity of the chills, hijacking our hardwiring for social bonding.

Imagine watching a tense scene in Jurassic Park or a magic trick that defies logic. The uncertainty of the situation could cause **anxiety** (a fear of the unknown). However, if the stimulus is aesthetically appreciated, the chill response acts as a "valve". Instead of retreating in anxiety, the aesthetic appreciation signals that this uncertainty is an **opportunity to learn**. The emotional state flips from anxiety to **curiosity**. The chill signals that the brain is successfully updating its internal models of the world, making the unknown feel exciting rather than threatening. This mechanism is so potent that listening to chill-inducing music can make people more willing to risk electric shocks just to satisfy their curiosity about a magic trick.

You are listening to a favorite song. It makes you feel energetic, your skin tingles, and you check your arm to see goosebumps. This is a chill: a marker of physiological arousal (increased heart rate and skin conductance) and is essentially a reward signal.

Conversely, if the song makes you feel physically calmer, your breathing slows down, and you feel a lump in your throat or weep, this is a tear response. While chills are akin to the "shivers" of energy and mixed happy/sad **euphoria**, tears are closer to a **cathartic** release involving physiological calming. A song that induces chills is often perceived as both happy and sad, while a song that induces tears is perceived primarily as sad and calm.

# Schlieren Photography

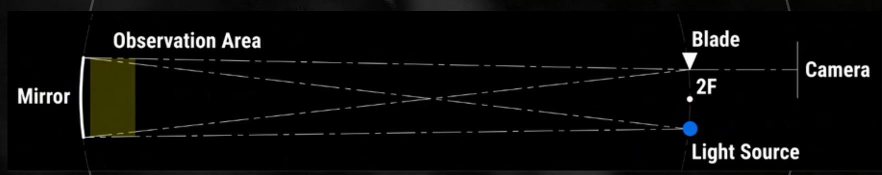
When lighting a candle, your naked eye can see the glowing yellow wick and perhaps a faint shimmer above it. But if you were to look at that same candle through a schlieren system, the empty space above the flame would erupt into a turbulent, rising column of greyscale or color. You would see the invisible plume of hot air fighting against the cooler room air, swirling like oil mixing with water.

This phenomenon occurs because light does not travel at a constant speed through our atmosphere; it changes speed based on the density of the air it passes through.

The "most important stuff" to understand about schlieren photography is that it visualizes refraction—the bending of light rays.

*The Physics:* The refractive index of air changes with temperature and density. Hot air (like that above a candle) is less dense than cold air. When light passes from cool air into hot air, it speeds up and bends. A schlieren system detects these tiny bends in light rays, converting them into changes in brightness or color that our eyes can see.

Traditional schlieren photography, developed in the 19th century by August Toepler, is an ingenious optical trick.




*Collimation:* A light source is shone through a slit and reflected off a high-quality concave mirror to create parallel light rays.

*The Disturbance:* These rays pass through the "test section" (where your candle or bullet is). If the air density is uniform, the light travels straight. If there is a shock wave or heat plume, the light rays deflect slightly.

*The Knife-Edge:* The light is focused back down to a point where a razor blade (knife-edge) is positioned to block exactly half the light.

If a light ray bends upward (due to heat), it might skip over the blade, making that part of the image bright. If a light ray bends downward, it hits the blade and is blocked, making that spot dark.

The Story of the Bullet

*In the 1930s, researchers used this method to photograph explosives and bullets. When a bullet screams through the air, it compresses the gas molecules ahead of it so violently that it creates a "surface of discontinuity"—a shock wave. To the schlieren camera, this shock wave appears as a sharp, defined line. Solid particles from the explosion might break through this wave, creating their own tiny conical shock waves, similar to the wake of a boat. This technique allowed scientists to see that these disturbances happen in mere millionths of a second.*

## Rainbow Schlieren: Coloring the Invisible

While standard schlieren gives you black-and-white images (intensity), Rainbow Schlieren gives you color (hue).

*How it works:* Instead of a metal knife-edge blocking the light, a radial rainbow filter (a bull's-eye pattern of colors) is placed at the focal point.

- *The Result:* Weak density changes might bend the light just enough to hit the blue ring of the filter. Stronger gradients (like a shock wave) might bend the light further, hitting the red ring.
- *The Benefit:* This turns the photograph into a quantitative map. You don't just see that the air is moving; the specific color tells you exactly how much the light was deflected.

## The Modern Evolution: BOS

Traditional schlieren requires massive, expensive mirrors and precise alignment. In the year 2000, a digital revolution occurred: Background Oriented Schlieren (BOS).

*Mechanism:* Instead of mirrors and knife-edges, BOS uses a digital camera and a textured background (like a sheet of random dots).

*Reference:* You take a photo of the dots with no flow (air at rest).

*Deflection:* You turn on the wind tunnel or light the fire. The density changes act like a weak lens, optically distorting the background pattern.

*Processing:* Computer algorithms (like cross-correlation or optical flow) compare the two images. They calculate exactly how many pixels the background dots "moved" due to the light bending.

The Story of the Jet Flame

When observing a jet flame with rainbow schlieren, the image becomes a varicolored map. The core of the flame might appear one color, while the mixing layers on the edges appear as different hues. This allows scientists to calculate the actual temperature distribution inside the flame based on the specific colors recorded.

The Story of the Helicopter

BOS is so flexible it can be taken out of the lab. Researchers have flown aircraft above a helicopter to photograph the invisible vortices trailing off the rotor blades. By using the natural background of the earth (like a quarry or gravel), they could calculate the density of the air simply by measuring how much the background rocks appeared to wiggle as the helicopter flew over them.